Teaching Statement

Maqsudur Rahman

Ph.D. Candidate in Computer Science, Boise State University, USA
Application: Lecturer - Information & Computer Science, Job Reference F109, KFUPM
Email: maqsudurrahman@u.boisestate.edu

Teaching Philosophy and Commitment

My teaching philosophy is grounded in three principles: **conceptual clarity**, **practice-based learning**, and **student-centered mentoring**. Computer science education is most effective when students understand both the theoretical foundation of a topic and its practical relevance to real computing systems. Therefore, I aim to teach core computing and artificial intelligence topics through a balanced approach that combines rigorous explanation, programming practice, laboratory activities, problem-solving, project development, and research-oriented thinking.

I am applying for the Lecturer - Information & Computer Science position at King Fahd University of Petroleum & Minerals because the role emphasizes high-quality Bachelor-level teaching, strong student interaction, and contribution to a department whose areas include information security, secure computing, software engineering, high-performance computing, bioinformatics, and information visualization. My teaching background, academic appointments, student supervision record, and current doctoral training in Computer Science align closely with these responsibilities.

Across my academic career, I have taught a broad range of undergraduate Computer Science and Information & Communication Technology courses, including **Structured Programming Language**, **Object-Oriented Programming Language**, **Data Structures**, **Algorithms**, **Operating Systems**, **Information Security**, **Artificial Intelligence**, **Machine Learning**, and **Deep Learning**. These courses have allowed me to teach students at different stages of academic development: from first-year programming foundations to advanced AI and security-related topics.

Teaching Experience and Course Preparation

Before beginning my Ph.D. at Boise State University, I served in full-time academic positions as Assistant Professor, Lecturer, and Senior Lecturer in computer science and ICT departments. These roles gave me extensive experience in undergraduate teaching, course delivery, laboratory instruction, assessment design, academic advising, project supervision, and student mentoring.

My course preparation strategy is outcome-driven. For each course, I first identify the core competencies students should acquire, then design lectures, assignments, labs, and assessments around those outcomes. For example, in programming and object-oriented programming courses, I emphasize code readability, debugging discipline, modular design, and incremental development. In data structures and algorithms, I connect mathematical reasoning with implementation, complexity analysis, and real-world performance trade-offs. In operating systems, I focus on process management, memory, concurrency, file systems, and the relationship between system-level concepts and modern computing environments. In information security, I emphasize secure design, threat awareness, cryptographic foundations, network and system vulnerabilities, and responsible computing practice.

For AI, machine learning, and deep learning courses, I combine conceptual foundations with hands-on implementation. Students need to understand not only how to run models, but also how to reason about data quality, model evaluation, overfitting, uncertainty, reproducibility, and ethical deployment. My current Ph.D. research in trustworthy AI, secure/adversarial machine learning, robust medical AI, and hybrid quantum-classical learning further strengthens my ability to connect AI education with emerging research and real applications.

Pedagogical Approach

My classroom practice is based on active learning and progressive skill development. I try to move students from passive familiarity to independent problem-solving. A typical teaching sequence in my courses involves: (i) introducing the key idea, (ii) explaining the formal or algorithmic structure, (iii) demonstrating examples, (iv) assigning guided practice, and (v) asking students to apply the idea in a programming task, case study, or mini-project.

I place particular emphasis on the following methods:

- **Concept-to-code learning:** Students should be able to translate abstract concepts into correct and maintainable code. This is especially important for programming, data structures, algorithms, operating systems, and AI/ML courses.
- **Laboratory-based instruction:** Practical computing education requires guided lab sessions where students learn through implementation, debugging, experimentation, and reflection.
- **Problem-solving culture:** I encourage students to reason from first principles, compare alternative solutions, and justify design decisions rather than memorize isolated procedures.
- **Research-informed teaching:** In advanced courses, I introduce students to current developments in AI, security, trustworthy computing, biomedical data analysis, and high-performance computing so that they see how classroom knowledge connects to active research.
- **Inclusive student interaction:** I support students through clear expectations, timely feedback, advising, and encouragement, while maintaining academic standards and fairness.

Assessment, Feedback, and Student Learning

Assessment should measure both knowledge and competence. In my teaching, I use a combination of quizzes, programming assignments, written problem sets, laboratory tasks, presentations, exams, and course projects. For programming and systems courses, I value correctness, clarity, testing, documentation, and design quality. For AI and machine learning courses, I emphasize data preprocessing, model selection, evaluation metrics, interpretation of results, reproducibility, and responsible use of computational tools.

I believe feedback is one of the most important parts of teaching. Students often improve significantly when they understand why a solution is incorrect or incomplete, not simply that it is wrong. I therefore aim to provide feedback that is specific, constructive, and connected to learning outcomes. When possible, I also use milestone-based project evaluation so that students receive guidance before final submission rather than only after completion.

At KFUPM, I would be prepared to teach and support Bachelor-level courses in programming, object-oriented programming, data structures, algorithms, operating systems, information security, artificial intelligence, machine learning, software engineering, and related ICS courses. I would also be able to contribute to laboratory modules and project-based courses in secure computing, data analysis, AI systems, and applied machine learning.

Student Mentoring, Supervision, and Advising

A major part of my academic identity is student mentoring. During my teaching career, I supervised **40 undergraduate projects and 8 thesis students** in the Department of Computer Science at Port City International University and **16 undergraduate projects and 7 thesis students** in the Department of Information and Communication Technology at Comilla University. In total, I have supervised **56 undergraduate projects and 15 thesis students**. These experiences helped me develop a mentoring style that balances independence, structure, and accountability.

In project supervision, I guide students through problem selection, literature review, system design, implementation, evaluation, documentation, and presentation. I encourage students to choose meaningful computing problems and to think carefully about feasibility, data quality, methodology, and ethical implications. My supervision experience has covered areas such as machine learning, deep learning, natural language processing, cybersecurity, data analysis, and software development.

I have also contributed to extracurricular academic development. I served as coach for the CoU "Unpredictable 3207" ACM ICPC team, which became a back-to-back divisional champion. This experience strengthened my commitment to programming contest training, peer learning, and student confidence-building. Beyond formal courses, I value technical seminars, programming activities, student clubs, and research discussions as important parts of a healthy computer science learning culture.

Alignment with KFUPM and Future Teaching Contributions

The KFUPM Lecturer position emphasizes quality education, Bachelor-level teaching, and student interaction in the Information & Computer Science Department. I am well prepared to contribute to this mission because my teaching experience spans foundational computing, systems, security, and AI/ML. My academic background also aligns with several of the department's advertised areas, including information security, secure computing, software engineering, high-performance computing, bioinformatics, and information visualization.

If appointed, I would aim to contribute in four ways. First, I would deliver rigorous and well-organized undergraduate courses that strengthen students' fundamentals in computing. Second, I would support laboratory and project-based teaching, helping students move from theory to implementation. Third, I would mentor students in research-oriented and industry-relevant projects, particularly in AI, security, trustworthy computing, biomedical data analysis, and software systems. Fourth, I would contribute to curriculum development and academic service through course review, assessment design, student advising, and participation in departmental initiatives.

My long-term teaching goal is to help students become technically strong, ethically responsible, and intellectually independent computing professionals. I believe that KFUPM's strong academic environment and emphasis on quality education would allow me to contribute meaningfully to undergraduate teaching, student mentoring, and the broader academic mission of the Information & Computer Science Department.

Last updated: July 4, 2026.